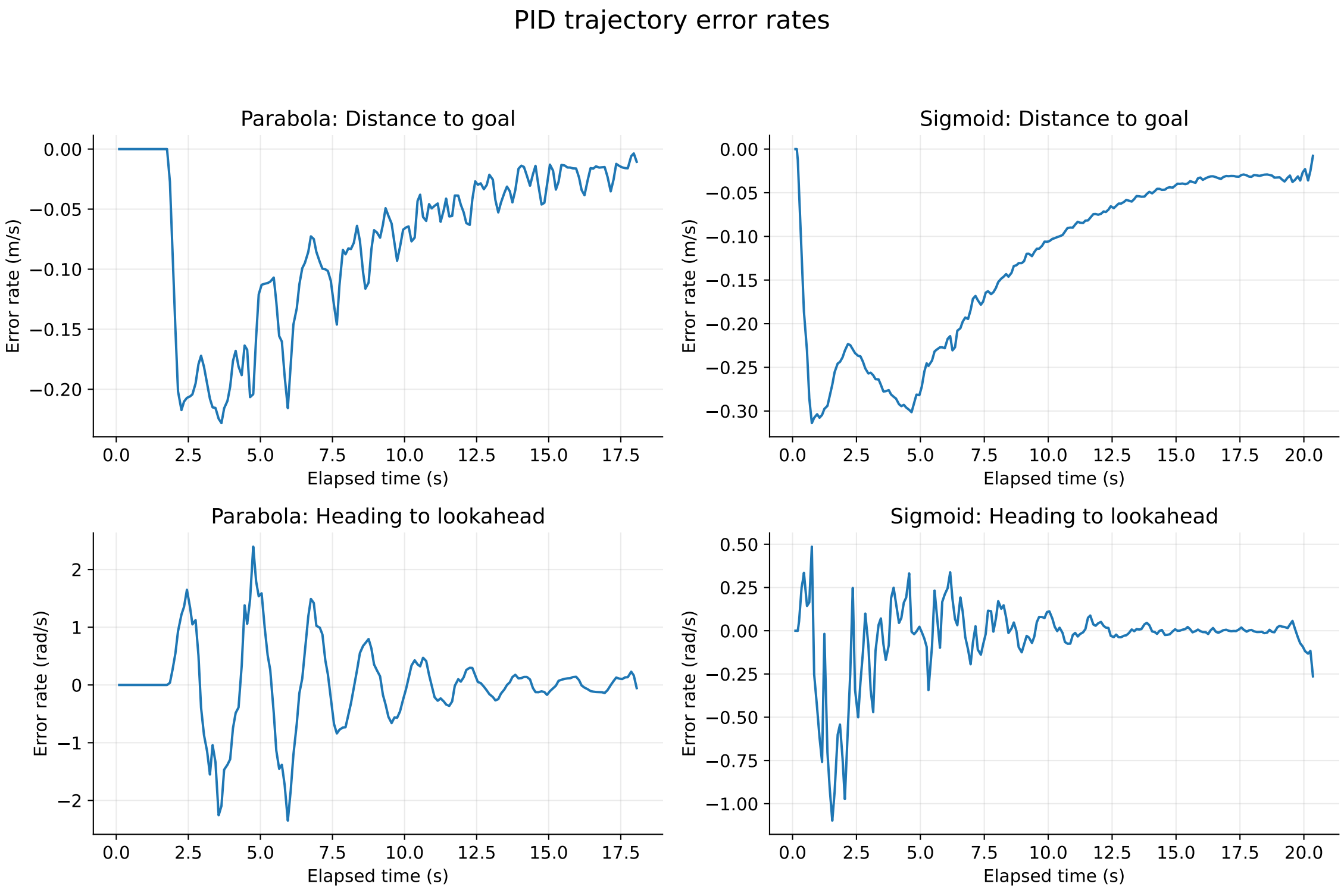
PID trajectory error rates
Parabola: Distance to goal
Error rate (m/s)
0.00
−0.05
−0.10
−0.15
−0.20
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Elapsed time (s)
Sigmoid: Distance to goal
Error rate (m/s)
0.00
−0.05
−0.10
−0.15
−0.20
−0.25
−0.30
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Elapsed time (s)
Parabola: Heading to lookahead
Error rate (rad/s)
2
1
0
−1
−2
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Elapsed time (s)
Sigmoid: Heading to lookahead
Error rate (rad/s)
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Elapsed time (s)